Course Outline for CSE-435

Part A

1. Course Code: CSE-435

2. Course Title: Information Security

3. Course Type: Core Course

4.Level/ Term: Level: 4 Term: I

5. Academic Session: 2020-21

6. Course Teacher: Dr. Md. Mokammel Haque, Professor, Dept. of CSE, CUET
Ashim Dey, Assistant Professor, CSE, CUET

7. Prerequisite(s): None

8. Credits: 3.00 (3 hours of class work per week)

9. Contact Hours: 3 hours of class work per week

10. Total Marks: 300

11. Rational of the Course:

This course covers knowledge about information security and different cryptographic primitives. It helps to understand various types of security threats and attacks. It also helps to gather basic knowledge of designing a secure cryptosystems to prevent such attacks. This is a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is

Course Content:

Concept and applications of cryptography; Techniques of designing efficient Cryptosystems: Symmetric key encryption- Stream ciphers, Block ciphers; Message authentication codes; Hash function; Digital signature; Asymmetric key encryption- Diffie-Hellman protocol, Trapdoor functions, RSA, Merkel puzzles, El-Gammal cryptosystems; Linear and differential cryptanalysis; Mathematics of cryptography; Lightweight cryptographic protocols; Steganography.
Fundamental properties and terminology of information security; System security management, analysis and control; Physical and logical security; Types of attacks;

Database security; Network security threats; WSN security; Fundamental of Cybersecurity; IoT network and IoT device security; Blockchain; Secure Cloud computing; Computer abuse; Legal and Ethical issues.

12. Course Objectives:

(a) To understand various types of security threats and prevention strategies.

(b) To learn how to design a cryptosystem and cryptanalyze or attack a cryptosystem.

(c) To learn ethical principles related to the computer, data and secure system.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

No.	Course Learning Outcomes (CLOs)	Bloom's Level (Optional)
CLO1	Understanding different types of threats and prevention strategies	
CLO2	Designing a cryptosystem and cryptanalyze a cryptosystem	
CLO3	Learning ethical principles pertained to computer and data.	

b) Mapping of CLOs with PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PL11	PL12
CLO1	X											
CLO2	X	X	X									
CLO3								X				

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

Course Plan

Lecture No	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Lesson 01 (week 01)	1. Overview of cryptography. 2. Fundamentals of information security 3. Simple classical cryptosystems 4. Cryptanalysis	Class Lecture	• Class test • Final Exam • Assignment	CLO2

Lesson 02 (week 01)	1. The big-O notation 2. Time estimates 3. P, NP, and NP-completeness 4. Shannon's theory 5. One-time pad	Class Lecture	• Class test • Final Exam • Assignment	CLO2
Lesson 03 (week 02)	1. Secure PRG 2. Stream ciphers (RC series, Salsa, CSS, etc.) 3. Cryptanalysis of Stream ciphers	Class Lecture	• Class test • Final Exam • Assignment	CLO2
Lesson 04 (week 02 and 03)	1. Block ciphers and secret key encryption 2. Description of DES 3. Description of AES 4. 2DES and 3DES analysis 5. Quantum cryptography	Class Lecture	• Class test • Final Exam • Assignment	CLO2
Lesson 05 (week 04)	1. Review of algebra and number theory: * Groups, fields, Rings * Euclidean algorithm for polynomials * Chinese remainder theorem * Other useful facts	Class Lecture	• Class test • Final Exam • Assignment	CLO2
Lesson 06 (week 04 and 05)	1. Message Authentication codes and their basic constructions (NMAC, ECBC MAC, PMAC, etc.) 2. Digital signatures	Class Lecture	• Class test • Final Exam • Assignment	CLO2
Lesson 07 (week 05 and 06)	1. Collision-free hash functions, Motivation, and applications, 2. Discrete-Log Hash function, 3. Merkle-Damgard	Class Lecture	• Class test • Final Exam • Assignment	CLO2

	and other constructions 4. MAC from collision resistance hashing			
Lesson 08 (week 06 and 07)	1. Key pre-distribution 2. Merkel puzzle and Diffie-Hellman key exchange	Class Lecture	• Class test • Final Exam • Assignment	CLO2
Lesson 09 (week 08)	1. Public-key cryptography concept 2. Practical PKC from Trapdoor functions 3. RSA cryptosystem and text-book RSA 3. ElGamal Crypto and PKCS V 1.0 4. Cryptanalysis on PKC	Class Lecture	• Class test • Final Exam • Assignment	CLO2
Lesson 10 (week 09)	1. Program and systems security	Class Lecture	• Class test • Final Exam • Assignment	CLO1
Lesson 11 (week 10)	2. Network Security	Class Lecture	• Class test • Final Exam • Assignment	CLO1
Lesson 12 (week 11)	3. Database Security	Class Lecture	• Class test • Final Exam • Assignment	CLO1
Lesson 13 (week 12 and 13)	1.Computer Ethics, cybercrime and cyber-security 2. Group presentation	Class Lecture	• Class test • Final Exam • Assignment	CLO3

Part C

15. Assessment and Evaluation

1) Assessment Strategy:

Quizzes	15%
Viva-voce	15%
Class performance including reports	60%
Attendance	10%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 70%

b) Summative: 30%

3) Make-up Procedures:

- Course teacher may arrange for makeup lab schedule if necessary.

Part D

16. Learning Materials

1) Recommended Readings

- Charles P. Pfleeger et al., Security in Computing, Prentice-Hall.
- Paar, Christof, et al, Understanding Cryptography, Springer

2) Supplementary Readings

- Josef Pieprzyk et al., Fundamentals of Computer Security, Springer.
- William Stallings, Cryptography and Network Security, Pearson Education.

3) Others

- Presentation slides